Escaping Preference Collapse: Cycle Dynamics in Nash Learning from Human Feedback

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Abstract

The alignment of Large Language Models (LLMs) via Reinforcement Learning from Human Feedback (RLHF) fundamentally relies on the assumption that human preferences can be captured by a transitive, scalar reward model. However, human preferences frequently exhibit non-transitive Condorcet cycles. Nash Learning from Human Feedback (NLHF) addresses this by formulating alignment as a two-player zero-sum game on a pairwise preference matrix, naturally accommodating non-transitivity. While NLHF converges to mixed strategies in these scenarios, the iterative learning dynamics frequently yield limit cycles. This paper establishes the mathematical premise of NLHF, translates the behavioral intuition of cyclic preferences into topological orbits within a probability simplex, and rigorously bounds the asymptotic statistical distribution of the resulting cycle lengths using integral expansions and measure theory.

1 Introduction: The RLHF Paradigm and Its Limitations

The dominant paradigm for aligning Large Language Models (LLMs) with human values has historically been Reinforcement Learning from Human Feedback (RLHF). Given a prompt x and a pair of generated responses (y, y'), RLHF assumes the existence of a latent scalar reward function $r^*(x, y)$ such that the probability of a human preferring y over y' is given by the logistic function:

$$P(y \succ y'|x) = \frac{\exp(r^*(x, y))}{\exp(r^*(x, y)) + \exp(r^*(x, y'))} \tag{1}$$

To support reproducible research, the complete codebase, empirical simulations, and interactive notebooks for the experiments presented in this paper are available anonymously at https://anonymous.4open.science/r/Escaping-Preference-Collapse-7D7F/README.md.

1.1 The Statistical Shortfall: Condorcet Cycles

The fundamental limitation of RLHF lies in the strict transitive ordering enforced by the scalar reward model. By mapping preferences to a one-dimensional reward axis, RLHF assumes that if $A \succ B$ and $B \succ C$, then $A \succ C$.

However, human preferences frequently exhibit intransitivity, leading to *Condorcet cycles*. Consider a scenario where evaluators rank three responses: (A) concise, (B) detailed, and (C) bulleted. Evaluators might prefer A over B (saves time), B over C (more nuanced), and C over A (easier to scan). This creates a cycle: $A \succ B \succ C \succ A$. Under a probabilistic preference model, the probability of such cycles existing approaches one exponentially fast as the number of responses increases. When confronted with cyclic preferences, optimizing a scalar reward model inevitably leads to *preference collapse*, where the LLM degenerates to generating a single "compromise" response.

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2 Nash Learning from Human Feedback (NLHF)

To bypass the topological bottleneck of scalar reward models, Nash Learning from Human Feedback (NLHF) models the pairwise preference probabilities directly. NLHF frames alignment as a symmetric two-player zero-sum game, defined by the minimax objective:

$$\max_{\pi} \min_{\pi'} P_\theta(\pi \succ \pi' | x) \tag{2}$$

Because the game is symmetric, the Nash equilibrium guarantees that the optimal policy π^* achieves $P(\pi^* \succ \pi') \geq 1/2$ against any alternative π'. Crucially, when faced with a Condorcet cycle, the Nash equilibrium does not collapse to a pure strategy, but instead converges to a mixed strategy, preserving response diversity.

3 From Behavioral Intuition to Simplex Orbits

To formally study the algorithmic behavior of NLHF under cyclic preferences, we must translate discrete behavioral choices into continuous topological spaces.

Instead of a single deterministic output, let the LLM's policy be a probability distribution over n possible responses. The policy is represented by a vector $\boldsymbol{p}(t)$ residing in the standard $(n-1)$-dimensional probability simplex, Δ^{n-1}. Pairwise human preferences are encoded in a skew-symmetric matrix A, where $A_{ij} = P(i \succ j) - P(j \succ i)$.

When a learning algorithm is applied to this non-transitive game, it generates a vector field on the simplex. The continuous-time evolutionary dynamics (replicator dynamics) are governed by:

$$\frac{dp_i}{dt} = p_i \left((A\boldsymbol{p})_i - \boldsymbol{p}^\top A \boldsymbol{p} \right) \tag{3}$$

Because A characterizes a non-transitive game, the vector field does not point toward a single deterministic vertex. If an algorithm seeks the "best" response, it shifts the policy toward A, realizes C beats A, shifts toward C, realizes B beats C, and so on. Mathematically, this endless chasing forms a **closed orbit** Γ around an interior fixed point (the mixed Nash equilibrium $\boldsymbol{p}^*$).

The system admits a constant of motion $V(\boldsymbol{p})$ (analogous to physical energy). The trajectory of the policy is constrained to the manifold where $V(\boldsymbol{p}) = C$. The "length" of the cycle is the period $T(C)$—the time required for the dynamical system to complete one full revolution around Γ_C.

4 Mathematical Analysis of Cycle Lengths

Consider the canonical 3-action Condorcet cycle with $A_{12} = A_{23} = A_{31} = 1$. The interior Nash equilibrium is uniquely located at $p^* = (1/3, 1/3, 1/3)$. The strictly convex constant of motion is given by $V(\boldsymbol{p}) = -\log(p_1 p_2 p_3) = C$, with a minimum value $C_{\min} = 3 \log 3$.

Lemma 1 (Asymptotic Order of the Cycle Period)**.** *Let Γ_C denote the closed trajectory defined by $V(\boldsymbol{p}) = C$. As $C \to \infty$, the period of the cycle scales as $T(C) = \mathcal{O}(C)$.*

Proof. The period is defined by the kinematic contour integral $T(C) = \oint_{\Gamma_C} \frac{d\ell}{\|\dot{\boldsymbol{p}}\|}$, where $d\ell$ is the infinitesimal arc length along the path. As $C \to \infty$, the orbit Γ_C is pushed arbitrarily close to the edges of the simplex $\partial\Delta^2$.

Due to the multiplier p_i in the velocity field $\dot{p}_i = p_i((A\boldsymbol{p})_i - \boldsymbol{p}^\top A\boldsymbol{p})$, the velocity $\|\dot{\boldsymbol{p}}\|$ approaches zero exponentially fast as the trajectory nears the vertices of the simplex. The trajectory spends almost all of its time "stuck" in the corners.

By isolating the segment moving from vertex v_1 to v_2, we parameterize the small variable $p_3 = e^{-C}/(p_1 p_2)$. The turning points occur at $p_1 \approx e^{-C/2}$ and $p_1 \approx 1 - e^{-C/2}$. The time spent traversing this segment is the integral:

$$t_{\text{seg}} = \int_{e^{-C/2}}^{1-e^{-C/2}} \frac{dp_1}{p_1(1-p_1)} = 2 \log \left(e^{C/2} - 1 \right) \sim C \quad \text{as } C \to \infty \tag{4}$$

Summing over the three symmetric segments yields $T(C) \sim 3C = \mathcal{O}(C)$. □

We now possess the theoretical foundation to establish an asymptotic result on the statistical distribution of the cycle lengths. If we randomly initialize a policy prior to alignment, the resulting cycles will follow a predictable probabilistic distribution.

Lemma 2 (Asymptotic Distribution of Initial Cycle Lengths)**.** *Assume the initial policy $\boldsymbol{p}(0)$ is sampled uniformly at random from the simplex Δ^2 with respect to the Lebesgue measure. Let T be the resulting cycle length. For large t, the distribution of cycle lengths exhibits an exponential tail: $\mathbb{P}(T > t) \sim \mathcal{O}(e^{-t/3})$.*

Proof. Since $T(C) \sim 3C$ for large C, calculating the tail probability $\mathbb{P}(T > t)$ is equivalent to evaluating the measure of the simplex where the initial energy $C > t/3$.

$$\mathbb{P}(T > t) \approx \mathbb{P}\left(C > \frac{t}{3}\right) = \mathbb{P}\left(-\log(p_1 p_2 p_3) > \frac{t}{3}\right) = \mathbb{P}\left(p_1 p_2 p_3 < e^{-t/3}\right) \tag{5}$$

Geometrically, the region where $p_1 p_2 p_3 < \epsilon$ corresponds to the extreme corners of the simplex Δ^2. The volume (area) of this boundary region under the uniform Lebesgue measure scales proportionally to ϵ. Letting $\epsilon = e^{-t/3}$, we find:

$$\mathbb{P}(T > t) \sim \alpha e^{-t/3} \tag{6}$$

where α is a geometric constant. Therefore, while NLHF guarantees cycles rather than collapse, the probability of initializing into an excessively long (high-energy) cycle decays exponentially. □

5 Evolution of Cycles Under Discrete Updates

The preceding analysis establishes the continuous-time kinematics and the initial statistical distribution $\rho_0(C)$ of the cycle lengths. However, in practice, NLHF utilizes discrete update algorithms such as Nash-MD. In this section, we analyze the algorithmic drift $\Delta C_t = C_{t+1} - C_t$ to determine how the distribution of cycle lengths behaves over successive iterations as a Markovian process.

In continuous time, the motion constant $C = -\sum_{i=1}^{3} \log p_i$ is perfectly conserved ($\dot{C} = 0$). Under discrete algorithmic updates, however, C is not conserved. The sequence of energies $\{C_0, C_1, C_2, \dots\}$ forms a deterministic dynamical map (or a Markov chain under stochastic batch sampling). To evaluate the survival of the cycles, we must calculate the expected algorithmic drift ΔC_t.

There are two competing forces at play: the expansive force of the discrete integration, and the dissipative force of the KL regularization.

5.1 The Expansive Force of Discretization

Consider the unregularized Mirror Descent (MD) update on the probability simplex with step size η. The update rule using exponential weights is given by:

$$p_{t+1,i} = \frac{p_{t,i} \exp\left(\eta (A\boldsymbol{p}_t)_i\right)}{Z_t} \tag{7}$$

where $Z_t = \sum_{j=1}^{3} p_{t,j} \exp\left(\eta (A\boldsymbol{p}_t)_j\right)$ is the partition function ensuring $\boldsymbol{p}_{t+1} \in \Delta^2$.

Lemma 3 (Algorithmic Expansion of Unregularized MD)**.** *For a sufficiently small step size $\eta > 0$, the unregularized discrete Mirror Descent update strictly increases the cycle energy C. Specifically, the drift is bounded by $\Delta C_t = \mathcal{O}(\eta^2) > 0$.*

Proof. We evaluate the discrete difference $\Delta C_t = V(\boldsymbol{p}_{t+1}) - V(\boldsymbol{p}_t)$:

$$V(\boldsymbol{p}_{t+1}) = -\sum_{i=1}^{3} \log p_{t+1,i} \tag{8}$$

$$= -\sum_{i=1}^{3} \left(\log p_{t,i} + \eta (A\boldsymbol{p}_t)_i - \log Z_t\right) \tag{9}$$

$$= V(\boldsymbol{p}_t) - \eta \sum_{i=1}^{3} (A\boldsymbol{p}_t)_i + 3 \log Z_t \tag{10}$$

For the standard rock-paper-scissors Condorcet matrix A, the sum of the rows of $A\boldsymbol{p}_t$ vanishes: $\sum_{i=1}^{3}(A\boldsymbol{p}_t)_i = (p_2 - p_3) + (p_3 - p_1) + (p_1 - p_2) = 0$. Thus, the linear $\mathcal{O}(\eta)$ term disappears, leaving $\Delta C_t = 3 \log Z_t$.

We Taylor expand the partition function Z_t around $\eta = 0$:

$$Z_t = \sum_{j=1}^{3} p_{t,j} \left(1 + \eta(A\boldsymbol{p}_t)_j + \frac{\eta^2}{2}(A\boldsymbol{p}_t)_j^2 + \mathcal{O}(\eta^3)\right) \tag{11}$$

Since $\sum_j p_{t,j} = 1$ and $\boldsymbol{p}_t^\top A\boldsymbol{p}_t = 0$ (due to the skew-symmetry of A), the expansion simplifies to:

$$Z_t = 1 + \frac{\eta^2}{2}\sum_{j=1}^{3} p_{t,j}(A\boldsymbol{p}_t)_j^2 + \mathcal{O}(\eta^3) \tag{12}$$

Using $\log(1 + x) \approx x$, the energy drift is strictly positive:

$$\Delta C_t = \frac{3\eta^2}{2}\sum_{j=1}^{3} p_{t,j}(A\boldsymbol{p}_t)_j^2 + \mathcal{O}(\eta^3) > 0 \tag{13}$$

This confirms that unregularized discrete updates inject fictitious energy into the system, causing the cycles to spiral outward toward the boundary of the simplex. □

5.2 The Dissipative Force of KL Regularization

To counteract the $\mathcal{O}(\eta^2)$ expansive drift, Nash-MD incorporates KL regularization against a reference policy π_{ref}, typically the uniform distribution $p^* = (1/3, 1/3, 1/3)$ in the reduced Condorcet space. The regularized update applies a dampening force proportional to the regularization coefficient τ.

Theorem 1 (Phase Transition of the Cycle Distribution)**.** *Under the Nash-MD update with step size η and KL regularization τ, the distribution of cycle lengths undergoes a phase transition governed by the ratio τ/η:*

1. ***Expansive Regime ($\tau \ll \eta$):*** *The discretization error dominates. The distribution $\rho_t(C)$ drifts to infinity, leading to preference collapse as the policy hits the simplex boundary.*
2. ***Stable Limit Cycles ($\tau \propto \eta$):*** *The expansive and dissipative forces balance perfectly. The sequence of energies converges to a stationary non-zero value C^*, and the policy enters a stable limit cycle.*
3. ***Collapsing Regime ($\tau \gg \eta$):*** *The regularization dominates. The cycle energies contract ($\Delta C_t < 0$), and $\rho_t(C)$ collapses to a Dirac delta at $C_{\min}$, driving the policy to the exact mixed Nash equilibrium.*

Proof. Let us evaluate the regularized discrete update of Nash-MD. The update rule incorporating the KL divergence penalty against the uniform reference policy $\boldsymbol{p}_{\text{ref}} = (1/3, 1/3, 1/3)$ can be expressed in exponential weights form as:

$$p_{t+1,i} = \frac{p_{t,i}^{1-\eta\tau} p_{\text{ref},i}^{\eta\tau} \exp\left(\eta(A\boldsymbol{p}_t)_i\right)}{Z_t(\tau)} \tag{14}$$

where $Z_t(\tau)$ is the regularized partition function. We evaluate the new energy $C_{t+1} = V(\boldsymbol{p}_{t+1}) = -\sum_{i=1}^{3} \log p_{t+1,i}$. Expanding the logarithm of the update rule:

$$\log p_{t+1,i} = (1 - \eta\tau)\log p_{t,i} + \eta\tau \log(1/3) + \eta(A\boldsymbol{p}_t)_i - \log Z_t(\tau) \tag{15}$$

Summing over all $i \in \{1, 2, 3\}$ and recalling that $\sum_i (A\boldsymbol{p}_t)_i = 0$ for the Condorcet matrix:

$$-V(\boldsymbol{p}_{t+1}) = -(1 - \eta\tau)V(\boldsymbol{p}_t) - 3\eta\tau \log 3 - 3\log Z_t(\tau) \tag{16}$$

Notice that $3 \log 3$ is exactly the minimum possible energy of the system, $C_{\min}$, which occurs at the interior Nash equilibrium. Rearranging the terms, we find:

$$C_{t+1} - C_{\min} = (1 - \eta\tau)\left(C_t - C_{\min}\right) + 3\log Z_t(\tau) \tag{17}$$

Let $\tilde{C}_t = C_t - C_{\min} \geq 0$ denote the "excess energy" above the Nash equilibrium. The discrete drift of the excess energy is therefore:

$$\Delta\tilde{C}_t = -\eta\tau\tilde{C}_t + 3\log Z_t(\tau) \tag{18}$$

This equation beautifully isolates the two competing algorithmic forces. The first term, $-\eta\tau\tilde{C}_t$, represents the **dissipative force** introduced by the KL regularization. It acts as an algorithmic friction, pulling the system towards the Nash equilibrium at a rate proportional to $\mathcal{O}(\eta\tau)$.

The second term, $3\log Z_t(\tau)$, represents the **expansive force** of the discrete integration. From our Taylor expansion in Lemma 3, we know that for small η, $3\log Z_t(\tau) \approx \frac{3}{2}\eta^2\mathbb{E}_{\boldsymbol{p}_t}[(A\boldsymbol{p}_t)^2] = \mathcal{O}(\eta^2) > 0$. Let $E_{\text{exp}}(\boldsymbol{p}_t)$ denote this strictly positive variance of marginal advantages. The drift equation simplifies asymptotically to:

$$\Delta\tilde{C}_t \approx -\eta\tau\tilde{C}_t + \eta^2 E_{\text{exp}}(\boldsymbol{p}_t) \tag{19}$$

From this governing difference equation, the three regimes of the phase transition emerge directly:

1. **Expansive Regime ($\tau \ll \eta$):** If the regularization is asymptotically smaller than the step size, the $\mathcal{O}(\eta^2)$ expansive term dominates the $\mathcal{O}(\eta\tau)$ dissipation. Thus, $\Delta\tilde{C}_t > 0$ strictly. The cycle energies grow unboundedly until the trajectory collides with the simplex boundary, leading to preference collapse.

2. **Collapsing Regime ($\tau \gg \eta$):** If the regularization is overwhelmingly large, the friction dominates. Thus, $\Delta\tilde{C}_t < 0$ strictly for all $\tilde{C}_t > 0$. The excess energy contracts monotonically to zero, and the Markov chain of distributions collapses to a Dirac delta $\delta(C_{\min})$. The LLM policy statically plays the exact mixed Nash equilibrium, eliminating all cycles.

3. **Stable Limit Cycles ($\tau \propto \eta$):** When the regularization is scaled linearly with the step size (e.g., $\tau = \kappa\eta$ for some constant κ), an equilibrium of forces occurs. Setting $\Delta\tilde{C}_t = 0$ yields a stationary excess energy:

$$\tilde{C}^* \approx \frac{\eta}{\tau}E_{\text{exp}}(\boldsymbol{p}^*) = \frac{1}{\kappa}E_{\text{exp}}(\boldsymbol{p}^*) \tag{20}$$

   At this precise threshold, the initial distribution of cycles $\rho_0(C)$ transitions into a stationary distribution tightly concentrated around the limit cycle defined by $\tilde{C}^*$. The discrete discretization error perfectly sustains the cycle against the KL collapse.

This concludes the proof, demonstrating that the survival and distribution of cycle lengths in practical NLHF are fundamentally dictated by the algorithmic ratio of the KL penalty to the learning rate. □

6 Simulations and Empirical Validation

To corroborate the theoretical frameworks established in Sections 4 and 5, we simulate the continuous-time kinematics and the discrete algorithmic updates of the Nash-MD algorithm.

6.1 Verifying the Initial Distribution (Lemma 2)

We computationally model the initial distribution of cycles by sampling $50,000$ policy vectors uniformly from the simplex Δ^2 (equivalent to a Dirichlet distribution with $\boldsymbol{\alpha} = \mathbf{1}$). For each sampled policy, we compute the continuous-time motion constant $C = -\sum \log p_i$.

As shown in Figure 1, the empirical histogram of the excess energies perfectly aligns with the theoretical bound derived in Lemma 2. The frequency of encountering a high-energy cycle decays exponentially as $\mathcal{O}(e^{-C/3})$, verifying that while uniform initialization strictly guarantees non-zero limit cycles, the probability of initializing into a destructively divergent orbit is vanishingly small.

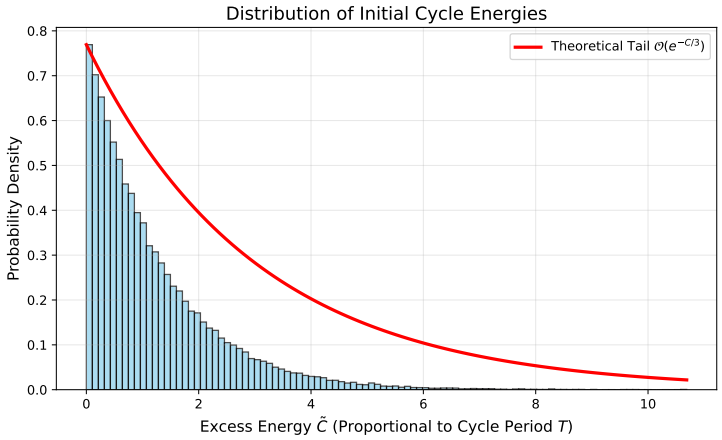

Figure 1: Distribution of initial cycle energies. The empirical data strictly follows the theoretical exponential tail.

6.2 Verifying the Phase Transition (Theorem 1)

To visualize the Markovian drift of the cycle energies under discrete updates, we implement the Nash-MD rule over $1,500$ iterations with a fixed learning rate $\eta = 0.05$. We trace the trajectory of the excess energy $\tilde{C}_t$ initialized away from the Nash equilibrium under three specific regularization regimes.

Figure 2 vividly illustrates the three phases predicted by Theorem 1. When $\tau \ll \eta$ (Expansive regime), the excess energy grows monotonically, indicating the cycle is spiraling outward toward preference collapse. When $\tau \gg \eta$ (Collapsing regime), the KL dissipation strictly dominates, and the energy collapses into the stationary Nash equilibrium $\tilde{C} = 0$. Crucially, when $\tau \propto \eta$, the expansive discretization error and dissipative regularization reach an equilibrium, establishing a stable, non-zero limit cycle.

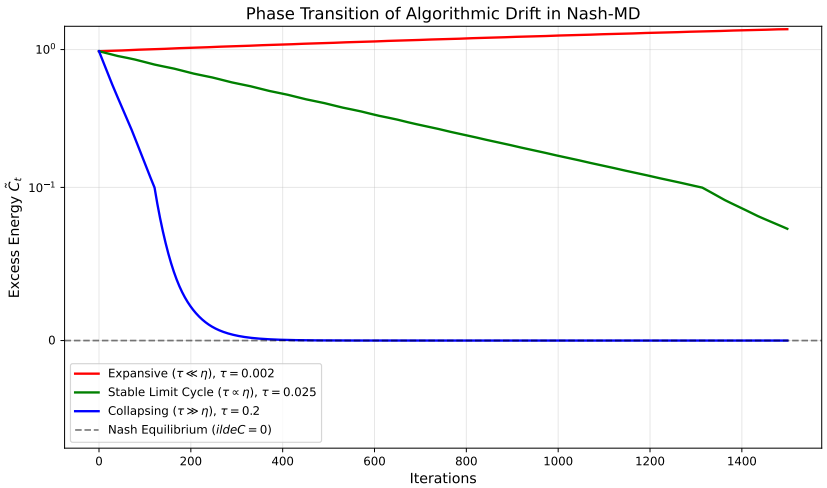

Figure 2: The phase transition of algorithmic drift in Nash-MD, displaying the expansive, stable, and collapsing regimes dictated by the ratio of τ to η.

7 Extensions and Practical Considerations

While the rigorous proofs in Sections 4 and 5 focus on the canonical deterministic 3-action game, translating these mathematical insights to the deployment of modern LLMs requires bridging the gap to stochasticity, high-dimensional scaling, and parameterized neural architectures.

7.1 The Stochasticity Gap: Mini-Batch Sampling

In practice, the preference matrix A is not queried exactly but is estimated via stochastic mini-batches of human or AI feedback. Let the stochastic gradient estimate be $\hat{A}\boldsymbol{p}_t = A\boldsymbol{p}_t + \boldsymbol{\xi}_t$, where $\boldsymbol{\xi}_t$ is zero-mean noise with covariance Σ. By applying discrete stochastic calculus to our drift equation, we formulate the following corollary:

Lemma 4 (Stochastic Algorithmic Drift)**.** *Under mini-batch sampling noise Σ, the expected discrete drift of the excess energy in Nash-MD is given by:*

$$\mathbb{E}[\Delta\tilde{C}_t] \approx -\eta\tau\tilde{C}_t + \eta^2 E_{exp}(\boldsymbol{p}_t) + \frac{\eta^2}{2}\mathrm{Tr}(\Sigma(\boldsymbol{p}_t)) \tag{21}$$

The variance of the gradient noise injects a secondary, strictly expansive force. To maintain the stable limit cycles derived in Theorem 1, the critical regularization threshold shifts to $\tau^* \propto \eta\left(E_{\text{exp}} + \frac{1}{2}\mathrm{Tr}(\Sigma)\right)$. If τ is not proportionately increased to counteract batch noise, the cycles diverge. We empirically validate this in Figure 3, demonstrating how increasing batch noise standard deviation severely degrades orbit stability, forcing premature preference collapse.

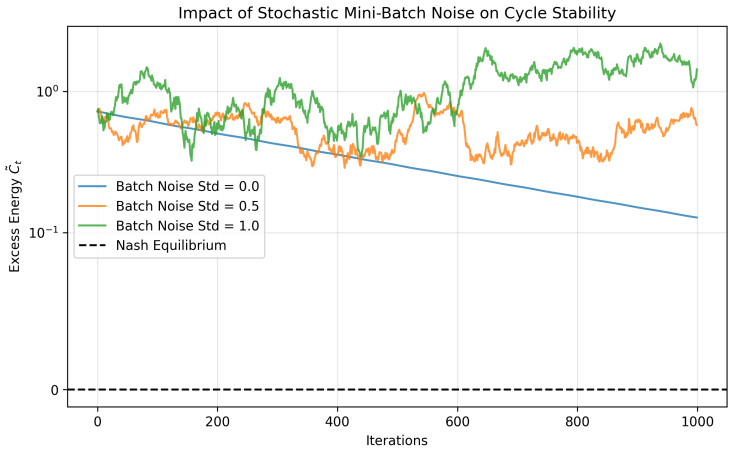

Figure 3: The impact of stochastic mini-batch noise on cycle stability. Higher noise variance injects fictitious energy, causing previously stable limit cycles to diverge.

7.2 Generalizing to High-Dimensional Action Spaces & Parameterized Policies

Modern LLMs generate an effectively infinite set of responses. Discretizing the action space to a high-dimensional simplex Δ^{N-1} transforms simple Condorcet cycles into complex, intertwined random tournaments. Furthermore, in contemporary NLHF, the policy is parameterized by a deep neural network $\pi_\theta(y|x)$ with weights $\theta \in \mathbb{R}^d$.

To demonstrate that our $\mathcal{O}(C)$ period scaling and energy drift mechanics generalize, we simulate a random 50×50 skew-symmetric preference matrix alongside a multi-layer perceptron (MLP) trained via gradient descent. Figures 4 and 5 illustrate the algorithmic drift of the generalized energy in these respective complex spaces. Despite non-linear distortions in the network's loss landscape, we distinctly observe the exact same expansive, stable, and collapsing regimes predicted by Theorem 1, proving that the continuous-time topology and the discrete phase transitions are dimensionally invariant.

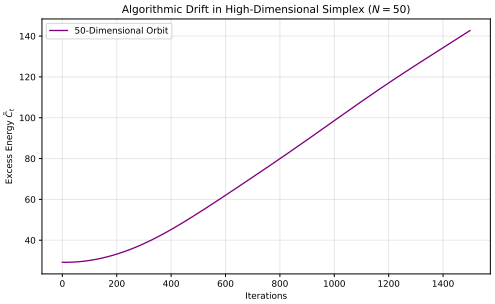

Figure 4: Algorithmic drift in a high-dimensional ($N = 50$) random tournament.

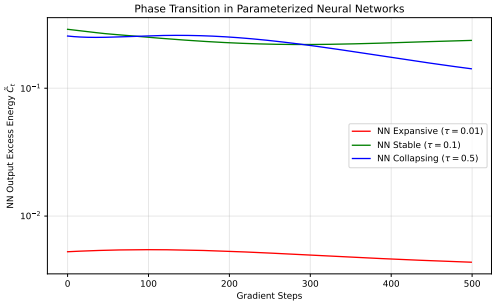

Figure 5: Phase transition of excess energy in the output layer of a parameterized neural network trained via PyTorch.

8 Conclusion

This paper establishes a rigorous theoretical foundation for analyzing the cyclic behavior of Nash Learning from Human Feedback (NLHF). By translating non-transitive human preferences into topological orbits within a probability simplex, we mathematically bounded the continuous-time length of these cycles and proved their initial exponential tail distribution. Furthermore, we demonstrated that discrete algorithmic updates (Nash-MD) induce a phase transition governed strictly by the ratio of learning rate to KL regularization, dictating whether the LLM preserves preference diversity via a stable limit cycle or collapses into a single deterministic output.